Chapter II.2-3, Week 3

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2.14. (a) Let S be a graded ring. Show that $\operatorname{Proj} S = \emptyset$ if and only if every element of S_+ is nilpotent.

(b) Let $\varphi : S \to T$ be a graded homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \operatorname{Proj} T \ : \ \mathfrak{p} \not\supset \varphi(S_+)\}$. Show that U is an open subset of $\operatorname{Proj} T$, and show that φ determines a natural morphism $f : U \to \operatorname{Proj} S$.

(c) The morphism f can be an isomorphism even when φ is not. For example, suppose that $\varphi : S_d \to T_d$ is an isomorphism for all $d \geqslant d_0$, where d_0 is any integer. Then show that $U = \operatorname{Proj} T$ and the morphism $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is an isomorphism.

Proof. (a) Denote $\operatorname{nil}' S := \cap_{\mathfrak{p}} \mathfrak{p}$, where $\mathfrak{p}$ ranges over all homogeneous prime ideals of S. It suffices to show that $\operatorname{nil}' S = \operatorname{nil} S$.

For any prime ideal $\mathfrak{p}$, the homogenization $h(\mathfrak{p}) \subset \mathfrak{p}$ generated by its homogeneous elements is clearly a homogeneous prime ideal. Then we have

$$\operatorname{nil}' S \supset \operatorname{nil} S = \bigcap_{\mathfrak{p}} \mathfrak{p} \supset \bigcap_{\mathfrak{p}} h(\mathfrak{p}) = \operatorname{nil}' S,$$

which gives us $\operatorname{nil}' S \subset \operatorname{nil} S$. We conclude that $\operatorname{nil} S = \operatorname{nil}' S$. Now, $\operatorname{Proj} S = \emptyset$ iff $S_+ \subset \operatorname{nil}' S = \operatorname{nil} S$, which is exactly our result.

(b) Let $\mathfrak{p} \in \operatorname{Proj} T$ be a homogeneous prime ideal. Then $\varphi^{-1}(\mathfrak{p})$ is a homogeneous prime ideal, and $\varphi^{-1}(\mathfrak{p}) \supset S_+$ implies

$$\mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(S_+),$$

i.e. $\mathfrak{p} \notin U$. Conversely, $\mathfrak{p} \supset \varphi(S_+)$ implies

$$\varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(S_+)) \supset S_+,$$

so that $\varphi^{-1}(\mathfrak{p}) \notin \operatorname{Proj} S$. We conclude that U is the inverse image of $\operatorname{Proj} S$ under the mapping of homogeneous prime ideals $\mathfrak{p} \mapsto \varphi^{-1}$. Hence, we have a natural map $f : U \to \operatorname{Proj} S : \mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$.

Fix a homogeneous $g \in T_+$. Then

$$\varphi^{-1}(\mathfrak{p}) \ni g \implies \mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \ni \varphi(g) \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(g)) \ni g$$

gives us that

$$f^{-1}(D_+(g)) = D_+(\varphi(g)) \subset U.$$

Hence, $f : U \to \operatorname{Proj} S$ is continuous. We must now define the map $f^\sharp : \mathcal{O}_S \to f_*(\mathcal{O}_T|_U)$ of sheaves.

It suffices to define $f^\sharp$ on basic open sets. Fix a homogeneous $g \in S_+$, and note that

$$f_*(\mathcal{O}_T|_U)(D_+(g)) = \mathcal{O}_T(f^{-1}(D_+(g)) \cap U) = \mathcal{O}_T(D_+(\varphi(g))) \cong T_{(\varphi(g))}.$$

Recall as that φ is degree preserving, so each of its localizations are as well. In particular, $\varphi_g : S_g \to T_g$ restricts to a map $\varphi_g : S_{(g)} \to T_{(\varphi(g))}$. Then the following commutative diagram gives us our sheaf map on $D_+(g)$.

$$\begin{tikzcd}
\mathcal{O}_S(D_+(g)) \arrow[r, "f^\sharp(D_+(g))"] \arrow[d, "\wr"] & f_*(\mathcal{O}_T|_U)(D_+(g)) \arrow[d, "\wr"] \\
S_{(g)} \arrow[r, "\varphi_g"'] & T_{(\varphi(g))}
\end{tikzcd}$$

To show that these maps glue together, we note that $D(g) \cap D(h) = D(gh)$, and consider the following commutative diagram.

$$\begin{tikzcd}
S_{(g)} \arrow[r, "\varphi_g"] \arrow[dr] & S_{(\varphi(g))} \arrow[dr] & \\
& S_{(gh)} \arrow[r, "\varphi_{gh}"] & S_{(\varphi(gh))} \\
S_{(h)} \arrow[r, "\varphi_h"'] \arrow[ur] & S_{(\varphi(h))} \arrow[ur] &
\end{tikzcd}$$

This defines our sheaf homomorphism on all of $\operatorname{Proj} S$, so that we're done.

(c) Fix d_0 such that $\varphi : S_d \to T_d$ is an isomorphism for all $d \geqslant d_0$. We wish to show that $U = \operatorname{Proj} T$, so it suffices to show that $\sqrt{\varphi(S_+)} = T_+$. One inclusion is clear since φ is degree preserving. Surjectivity of $\varphi : S_d \to T_d$ gives us $\varphi(S_+) \supset \oplus_{d \geqslant d_0} T_d$. Then, for any element $a \in T_+$, a^{d_0} is of degree at least d_0, so $a^{d_0} \in \varphi(S_+)$ implies $a \in \sqrt{\varphi(S_+)}$. We conclude that $\sqrt{\varphi(S_+)} = T_+$, so $U = \operatorname{Proj} T$.

It suffices to show that $f^\sharp$ is an isomorphism on open sets $D(g)$. Fix homogeneous $g \in S_+$. Since $D(g^n) = D(g)$ for all $n > 0$, we may assume g is of degree $\geqslant d_0$. We show the map $\varphi_g : S_{(g)} \to T_{(\varphi(g))}$ is surjective. For any $b/\varphi(g)^n \in (B_A)_{\mathfrak{p}}$, $b\varphi(g) = \varphi(a)$ with $a \in A$ gives us

$$\varphi_g(a/g^{n+1}) = \varphi(ag)/\varphi(g^{n+1}) = b/\varphi(g)^n \in \operatorname{im} \varphi_g.$$

It suffices to check injectivity. If $\varphi(a/g) = 0$, then we can find $n \geqslant 1$ such that $\varphi(g)^n \varphi(a) = 0$. Then $\varphi(g^n a) = 0$ implies $g^n a = 0$ by injectivity of φ, whence $a/g = 0$ in $S_{(g)}$. The result follows. $\square$

2.18. In this exercise, we compare some properties of a ring homomorphism to the induced morphism of the spectra of the rings.

(a) Let A be a ring, $X = \operatorname{Spec} A$, and $f \in A$. Show that f is nilpotent if and only if $D(f)$ is empty.

(b) Let $\varphi : A \to B$ be a homomorphism of rings, and let $f : Y = \operatorname{Spec} B \to X = \operatorname{Spec} A$ be the induced morphism of affine schemes. Show that φ is injective if and only if the map of sheaves $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is injective. Show furthermore in that case f is *dominant*, i.e., $f(Y)$ is dense in X.

(c) With the same notation, show that if φ is surjective, then f is a homeomorphism of Y onto a closed subset of X, and $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective.

(d) Prove the converse to (c). Namely, if $f : Y \to X$ is a homeomorphism onto a closed subset, and $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective, then φ is surjective.

Proof. (a) The result is trivial, as $D(f) = \emptyset$ iff $f \in \cap_{\mathfrak{p}} \mathfrak{p} = \operatorname{nil} f$.

(b) In the language of coherent sheaves, we have $\mathcal{O}_X = \widetilde{A}$ and $f_*\mathcal{O}_Y = f_*\widetilde{B} = \widetilde{B_A}$. Then $f^\sharp_\mathfrak{p} : \mathcal{O}_{X,\mathfrak{p}} \to (f_*\mathcal{O}_Y)_\mathfrak{p}$ corresponds to the map $\varphi_\mathfrak{p} : A_\mathfrak{p} \to (B_A)_\mathfrak{p}$. By CA, we have that φ is injective iff each $\varphi_\mathfrak{p}$ is injective, which by the above occurs iff $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is injective.

It suffices to show that $f(Y)$ is dense in X. Fix $g \in A$. Then

$$\begin{aligned} f(Y) \cap D(g) = \emptyset &\iff f^{-1}(D(g)) = \emptyset \\ &\iff D(\varphi(g)) = \emptyset \\ &\iff \varphi(g) \in \operatorname{nil} B \\ &\iff g \in \operatorname{nil} A \\ &\iff D(g) = \emptyset \end{aligned}$$

by (a) and injectivity of φ respectively. Any nonempty open $U \subset X$ contains a nonempty distinguished open of X, so $f(Y)$ is dense in X.

(c) First, suppose that $\varphi : A \to B$ is surjective. Localization is exact, so each of the stalks $\varphi_\mathfrak{p} : A_\mathfrak{p} \to (B_A)_\mathfrak{p}$ is surjective, whence $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective. Surjectivity of φ gives us

$$\mathfrak{p} \supset \mathfrak{a} \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{a}) \implies \mathfrak{p} = \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(\varphi^{-1}(\mathfrak{a})) = \mathfrak{a},$$

so that

$$\begin{aligned} f(V(\mathfrak{a})) &= \{\varphi^{-1}(\mathfrak{p}) \in X \ : \ \mathfrak{p} \supset \mathfrak{a}\} \\ &= f(Y) \cap V(\varphi^{-1}(\mathfrak{a})). \end{aligned}$$

In particular,

$$f(Y) = f(Y) \cap V(\ker\varphi) \subset V(\ker\varphi).$$

Take $\mathfrak{q} \in V(\ker\varphi)$, and suppose there exists $a \in \varphi^{-1}(\varphi(\mathfrak{q})) \setminus \mathfrak{q}$. Then there is some $a' \in \mathfrak{q}$ with $\varphi(a) = \varphi(a')$, so that $a - a' \in \ker\varphi \setminus \mathfrak{q} = \emptyset$, a contradiction. Thus, $\varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$.

We now show that $\varphi(\mathfrak{q})$ is prime. Suppose $bb' \in \varphi(\mathfrak{q})$, and take $a, a' \in A$ such that $\varphi(a) = b$ and $\varphi(a') = b'$. Then $aa' \in \varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$ implies $a \in \mathfrak{q}$ or $a' \in \mathfrak{q}$. Hence, $b \in \varphi(\mathfrak{q})$ or $b' \in \varphi(\mathfrak{q})$ gives us that $\varphi(\mathfrak{q})$ is prime, so that $g : V(\ker\varphi) \to Y : \mathfrak{q} \to \varphi(\mathfrak{q})$ is a set-theoretic inverse to $f : Y \to V(\ker\varphi) \subset X$. In particular, $f(Y) = V(\ker\varphi)$.

It suffices to show that $g : V(\ker\varphi) \to Y$ is continuous. We write

$$f(V(\mathfrak{a})) = f(V) \cap V(\varphi^{-1}(\mathfrak{a})) = V(\ker\varphi + \varphi^{-1}(\mathfrak{a})),$$

so that $g^{-1}(V(\mathfrak{a})) = V(\ker\varphi + \varphi^{-1}(\mathfrak{a}))$ is closed in X. We conclude that f is a homeomorphism onto its closed image $f(Y) = V(\ker\varphi)$.

(d) Suppose now that $f : Y \to X$ is a homeomorphism onto a closed subset, and $f^\sharp : \mathcal{O}_X \to \mathcal{O}_Y$ is surjective. Let $Z = \operatorname{Spec}(A/\ker\varphi)$, so that the universal property of quotients supplies $\overline{\varphi} : A/\ker\varphi \to B$ such that $\varphi = \overline{\varphi} \circ \pi$. We denote the associated morphism of sheaves $\overline{f} : Y \to Z$. Note that $\pi : A \to A/\ker\varphi$ is surjective, so (c) gives us that the associated map $g : Z \to V(\ker\varphi) \subset X$ is a homeomorphism.

$\overline{\varphi}$ is injective, so (b) implies $\overline{f}^\sharp : \mathcal{O}_Z \to \mathcal{O}_Y$ is injective, and $\overline{f}(Y) = f(Y)$ is dense. $f(Y) = f(Y)^-$ is closed, so $\overline{f} : Y \to Z$ is surjective. $f = g \circ \overline{f}$ with f, g homeomorphisms onto their image implies $\overline{f}$ is a homeomorphism onto its image $\overline{f}(Y) = Z$. Additionally, $f^\sharp = \overline{f}^\sharp g^\sharp$ surjective implies $\overline{f}^\sharp$ is surjective, so that $\overline{f}^\sharp : \mathcal{O}_Z \to \mathcal{O}_Y$ is an isomorphism of sheaves. Then

$$\varphi = \overline{\varphi} \circ \pi = \overline{f}^\sharp(Z) \circ \pi$$

is the composition of a surjective map with an isomorphism, and is therefore surjective itself. □